JAGADISH GOSALA

Senior Full-Stack Engineer

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Hyderabad, India · Open to Remote, Hybrid or Relocation · Notice: currently serving, available to join 30 Nov 2026

SUMMARY

Full-stack engineer who leads a 6-engineer team and owns two production platforms at Provakil. Built a self-serve analytics platform — UI-composed filters and groupings compiled into MongoDB aggregation pipelines, executed asynchronously — and re-architected it from 2.5M to 15M records/day (~6x). Also built an organisation-wide LLM gateway with usage metering, PII masking, and chargeback, and shipped a multi-tenant SaaS product solo. Built and deployed a RAG legal research assistant over 11K Supreme Court judgments with hybrid retrieval, verified paragraph-level citations and measured evals. 5 years across backend, data, and product engineering.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL
Backend: Node.js, Express.js, Sails.js, FastAPI
Frontend: Vue.js, Next.js
Data: MongoDB (aggregation pipelines, index design), PostgreSQL, MySQL, Redis, Elasticsearch, Qdrant
AI & Retrieval: RAG, Hybrid search (dense + BM25, RRF), BGE-M3 embeddings, LLM-as-judge evaluation, Retrieval evaluation (MRR, hit@k), SSE streaming, Ollama, Presidio (PII masking)
AI Coding Agents & LLM APIs: Claude Code, Claude (agentic workflows), OpenAI Codex, GitHub Copilot, Gemini API, DeepSeek API, Groq, Cloudflare Workers AI
Infrastructure: Docker, RabbitMQ, AWS (S3, EC2), MinIO, Supabase, Vercel, Render, Neon, Upstash, WebSockets, Alembic, Cypress, Playwright, pytest, GitHub Actions

PROFESSIONAL EXPERIENCE

PROVAKIL — Software Developer, Team Lead **May 2023 – Present**
Enterprise legal-tech SaaS for banks and insurers

Platform Ownership & Engineering Leadership

- Lead a 6-engineer team owning the analytics platform end to end — architecture decisions, design reviews, code review, and on-call incident response. Interview candidates and mentor juniors through onboarding.
- Re-architected the pipeline from 2.5M to 15M records/day (~6x): pushed filter predicates into the join stage instead of filtering post-join, and added compound indexes derived from cross-collection access patterns after confirming write amplification on source collections was negligible.
- Replaced full-pipeline recomputation with incrementally-refreshed materialized views for the most-requested report types — a source-document update enqueues a single-record refresh, trading configurable dynamic fields for near-real-time freshness. Unblocked a client expansion of 200+ enterprise users.
- Partitioned processing into scheduled-refresh, email-delivery, and long-running (>1 hr) workload classes on independently scaled consumers, isolating slow aggregations so they no longer block routine refreshes.
- Own and extended a metadata-driven entity platform (AnyCabinets) where new object types — contracts, obligations, patents — are defined in stored configuration rather than code, comparable to Salesforce custom objects. Built a bank-specific contract-review and checklist workflow on top of it with AI-assisted extraction.
- Classified every inbound support ticket over six months to build a root-cause picture of the module, then eliminated the dominant failure classes in query logic and data sync — cutting ticket inflow ~70% (~15 ops-hours/week).
- Built the platform's Vue.js interface — filter builders, dynamic charts, and data tables driven by the same configuration model as the backend.

Backend Services & Infrastructure

- Built an organisation-wide LLM gateway (FastAPI) fronting every model call — wallet/ledger credit accounting, per-team and per-developer limits, PII masking before requests leave the network, provider routing, and rate-card chargeback. Ships as its own container for cloud and on-prem.
- Built an e-signature and statutory stamp-duty platform from scratch — provider integrations, document state machine, a dead-letter queue for failed requests, and environment-aware routing to S3 or MinIO.
- Built the customer-onboarding bulk-import pipeline — Python consumers coordinating via atomic Redis counters to emit throttled WebSocket progress events across parallel importers.

- Instrumented monitoring across the reporting stack — crashed-job detection, cron and generation-failure alerting, and Cypress regression coverage for the document pipeline.
- Built a production-to-on-prem migration tool reconciling cross-environment schema differences, enabling 3 enterprise on-prem deployments.

Applied LLM Systems

- Built a structured-output clause-rewrite tool that reads a contract together with its email correspondence and proposes redlines for review.
- Built multi-stage email automation — an LLM links inbound threads to the correct case record and chains modules so a conversation advances a matter from pre-litigation to litigation without manual data entry.
- Built a multi-language translation pipeline for 8+ languages including right-to-left scripts (Arabic, Hebrew, Urdu) — layout-direction switching, font fallbacks, and formatting preservation; cut per-document cost ~85% via prompt caching and batching.
- Built Python summarisation and metadata-extraction pipelines over documents running 50+ pages — removed 2–3 hours of manual reading per case and cut ops-team tagging by 50%+ (previously 40+ man-hours/day). Took an extraction proof-of-concept for daily court docket PDFs — inconsistent layouts, no fixed schema — into a production ingestion pipeline.

RJEYS INFINITY SOLUTIONS — Assistant Solutions Engineer **April 2021 – May 2023**

Software services firm

- Built backend APIs and production debugging workflows on a Node.js/Express service serving 10K+ daily active users, plus a client data-migration tool handling schema mapping, validation, and edge cases for 500K+ legacy records.
- Served as technical analyst for Gainsight; rewrote heavy automated batch jobs against MySQL with optimised queries and indexing — cut runtimes by 60%.

PROJECTS

LEGAL CASE RESEARCH ASSISTANT (RAG) — sole engineer **legal-case-research-assistant.vercel.app**

- Built and deployed a RAG research assistant over 11,018 Indian Supreme Court judgments that cites the exact judgment and paragraph. PDF ingestion (PyMuPDF) strips reporter margin letters by geometry, splits headnotes from the judgment and follows OCR-tolerant paragraph numbering: numbered-paragraph detection 66% → 98.2%; 432,881 chunks indexed.
- Hybrid retrieval in FastAPI over Qdrant (BGE-M3 dense + BM25 sparse, reciprocal rank fusion) with a two-stage citation mode: pick judgments on editor headnotes, then paragraphs within them. A provider interface swaps local (Ollama) and hosted inference by configuration.
- Streaming cited answers (Groq gpt-oss-120b, falling back to Workers AI Llama 3.3 70B only before the first token), every [n] citation verified against the excerpts: citation precision 1.0, all off-corpus questions refused, LLM-judged faithfulness 0.70 → 93% of answers fully faithful.
- Case analysis: an LLM plans issues and queries from the user's facts, multi-query results are fused with RRF, and a second call writes a cited research note (similar cases with outcomes for the user's side, arguments accepted and rejected), streamed over SSE.
- Golden set and eval harness (hit@k, MRR, paragraph hit, latency, LLM-judged faithfulness): on the full corpus the right judgment is in the top 5 for 92% of questions (MRR@10 0.75), paragraph hit@3 0.29 → 0.43, p50 latency 375 ms → 37 ms; dropped the reranker after it showed no gain at 4.6 s per query.
- Runs at ₹0/month on free tiers (Vercel, Render, Qdrant Cloud, Neon, Upstash): Redis quotas, answer cache and rate limits (fail-open), Alembic migrations, JSON logs, GitHub Actions CI with strict mypy, 100 pytest tests and Playwright e2e.

CAMPUS PLACEMENT PLATFORM — sole engineer **hyrd.co.in**

- Built a multi-tenant SaaS platform for university recruiting — four role types with hierarchical admin sub-roles, access control enforced through Postgres row-level security policies rather than in application code, and verified tenant isolation.
- Built an AI resume-tailoring workspace that scores a resume against a specific job description and company profile, returns per-section findings with reasoning, and generates a rewrite presented as a live diff — all under explicit anti-fabrication constraints and per-user rate limits.
- Shipped bulk CSV/Excel import with validation, magic-link invites, audit logging, web-push/PWA notifications and PDF/Excel reports on Next.js, TypeScript, PostgreSQL (Supabase) and Vercel.

EDUCATION

B.Tech, Electronics and Communications Engineering — RGUKT **2017 – 2021**